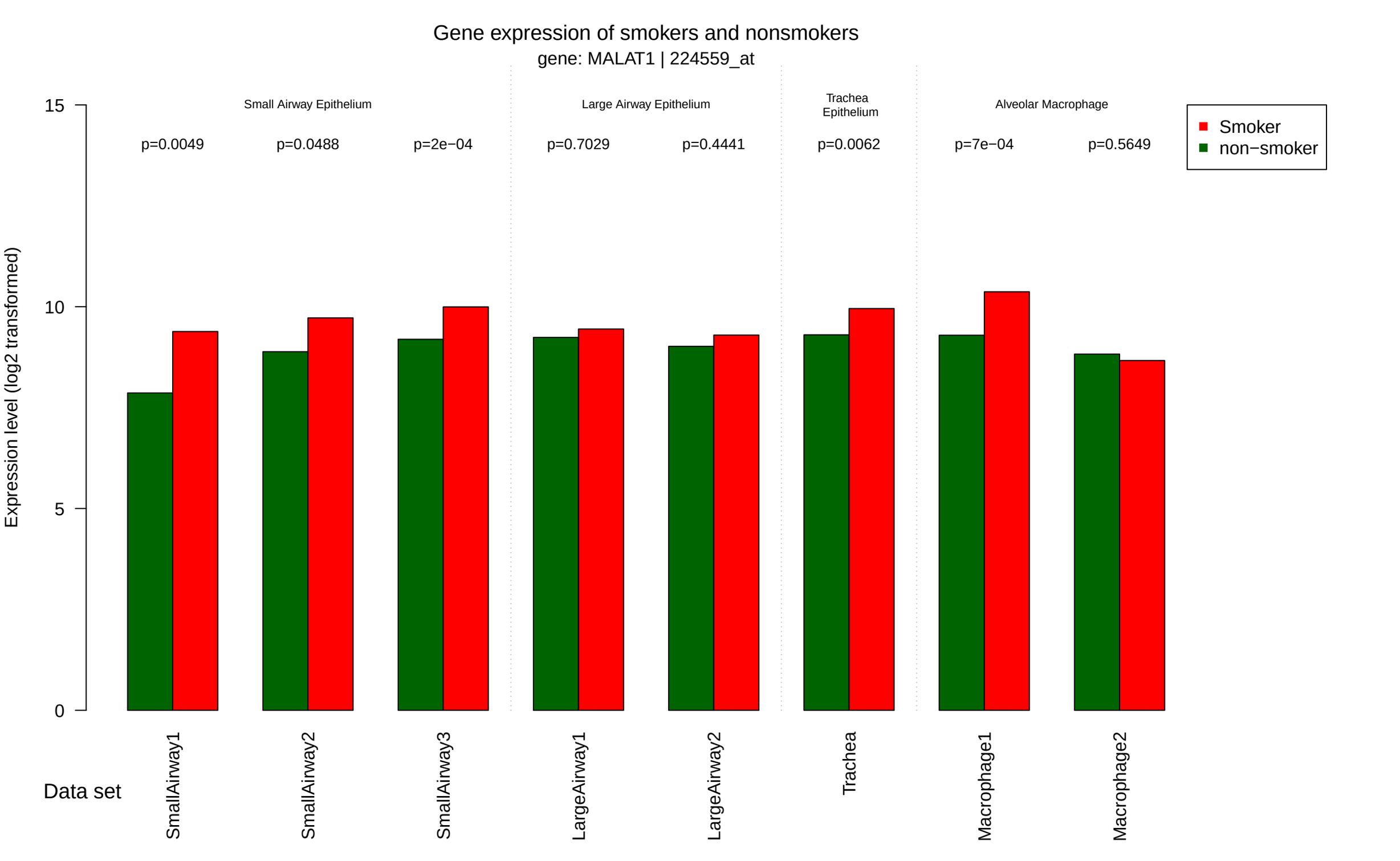
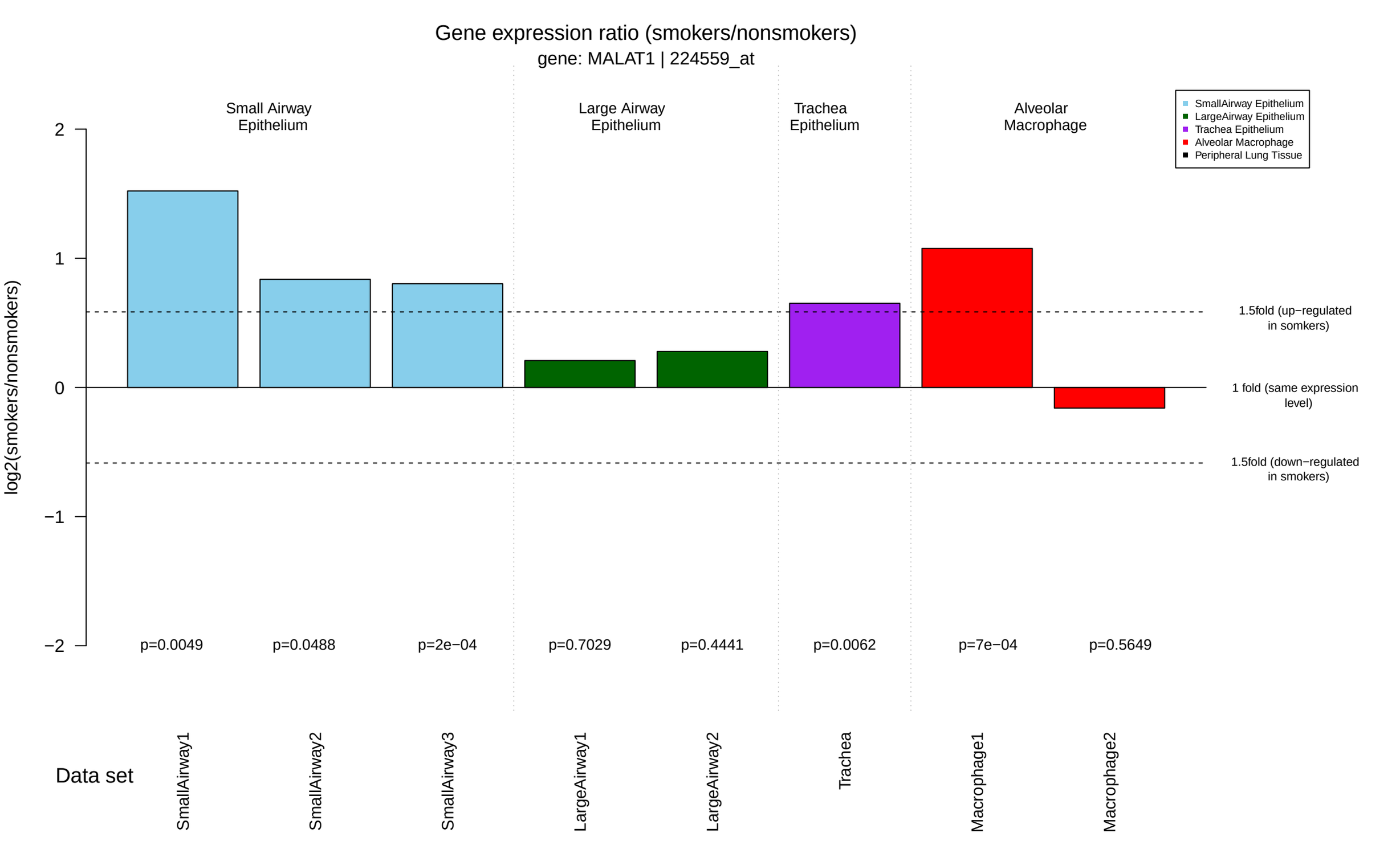
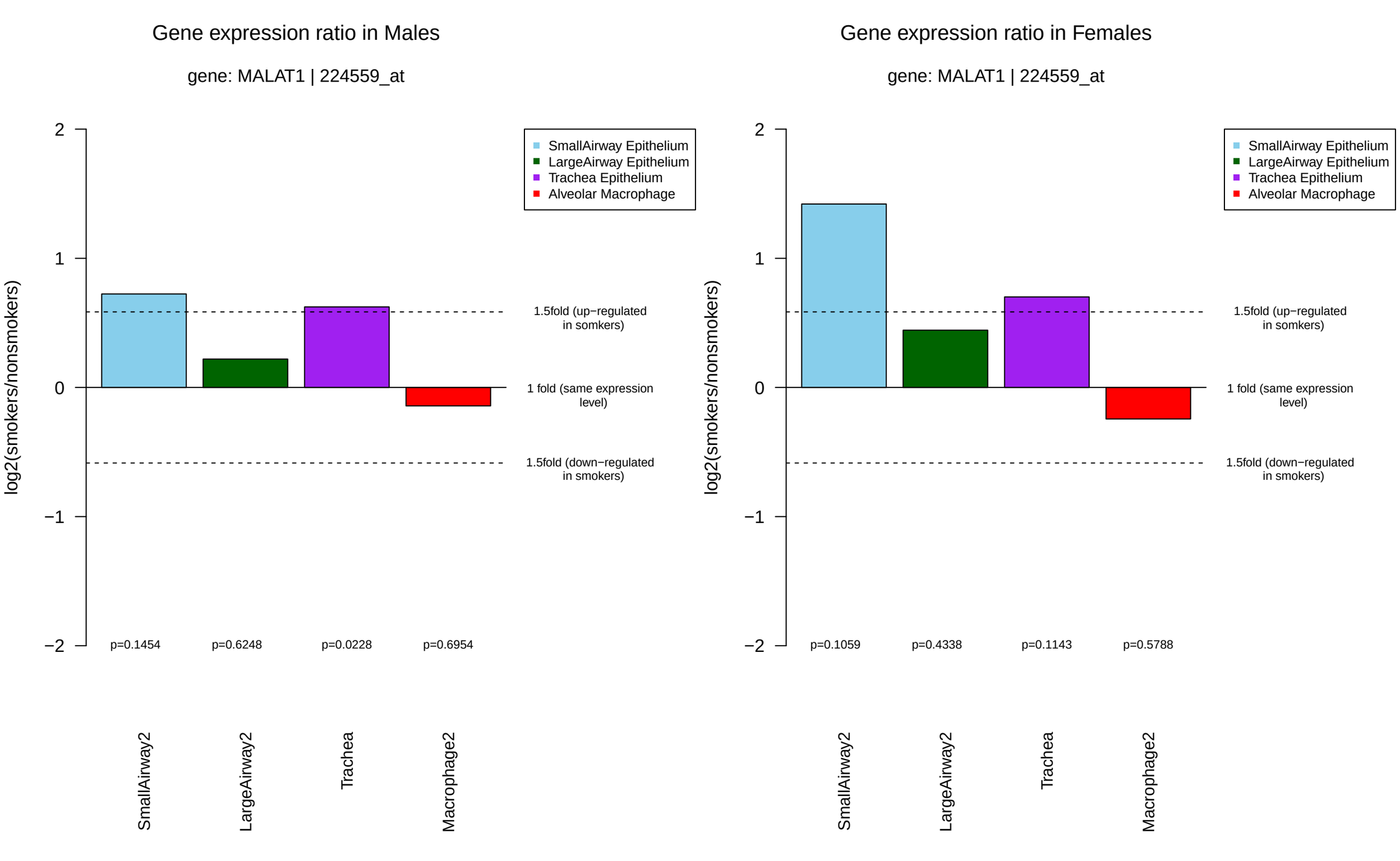
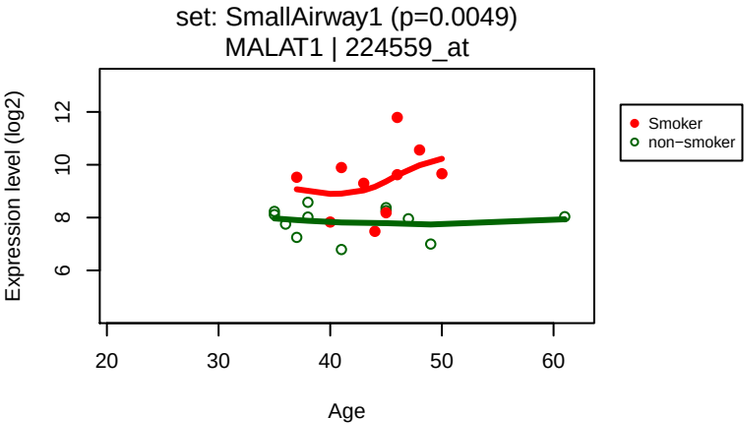
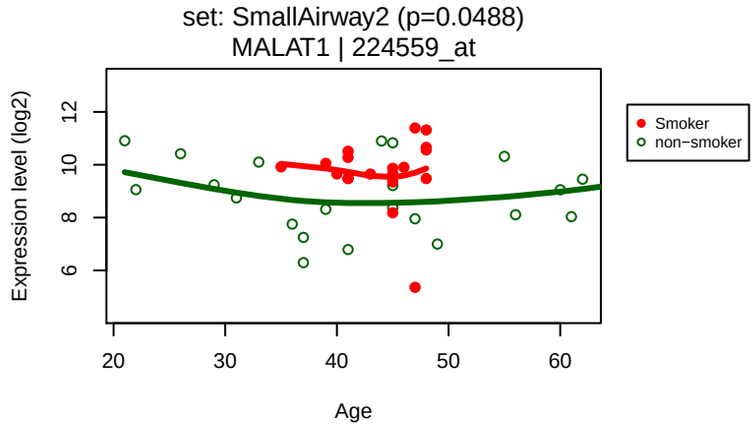
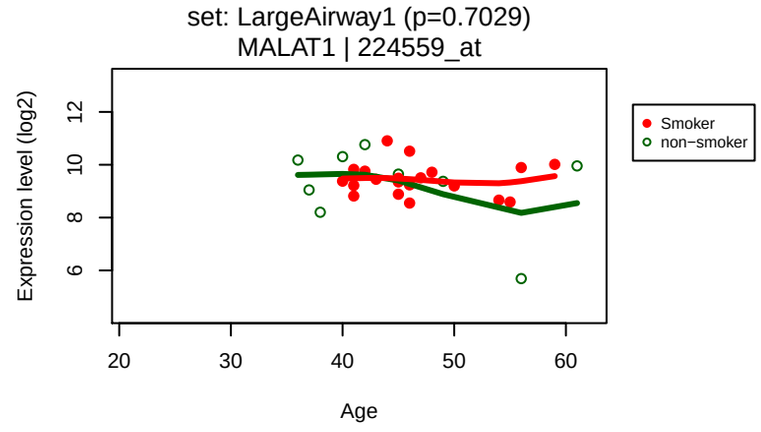

Gene expression of smokers and nonsmokers
gene: MALAT1 | 224559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0049
p=0.0488
p=2e−04
p=0.7029
p=0.4441
p=0.0062
p=7e−04
p=0.5649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

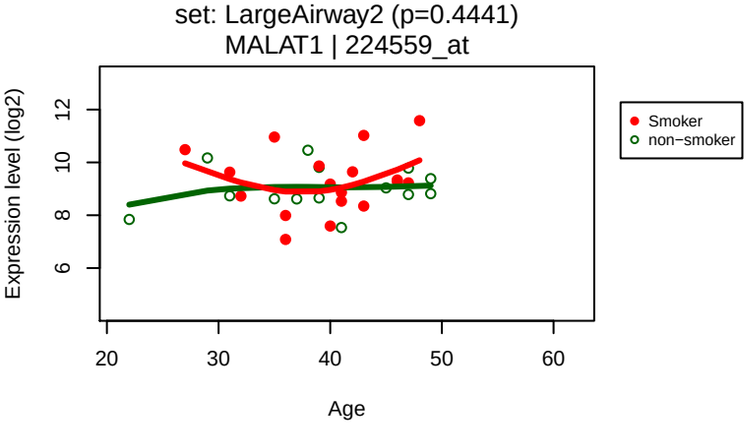

Gene expression ratio (smokers/nonsmokers)
gene: MALAT1 | 224559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=0.0488
p=2e−04
p=0.7029
p=0.4441
p=0.0062
p=7e−04
p=0.5649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

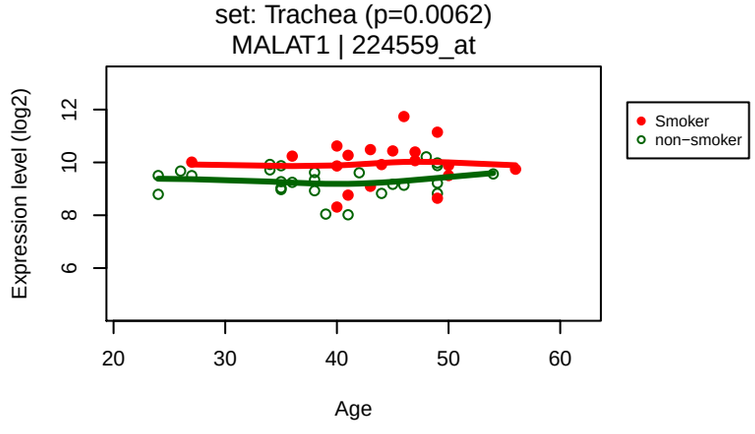

Gene expression ratio in Males
gene: MALAT1 | 224559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1454
p=0.6248
p=0.0228
p=0.6954
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MALAT1 | 224559_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1059
p=0.4338
p=0.1143
p=0.5788
SmallAirway2
LargeAirway2
Trachea
Macrophage2

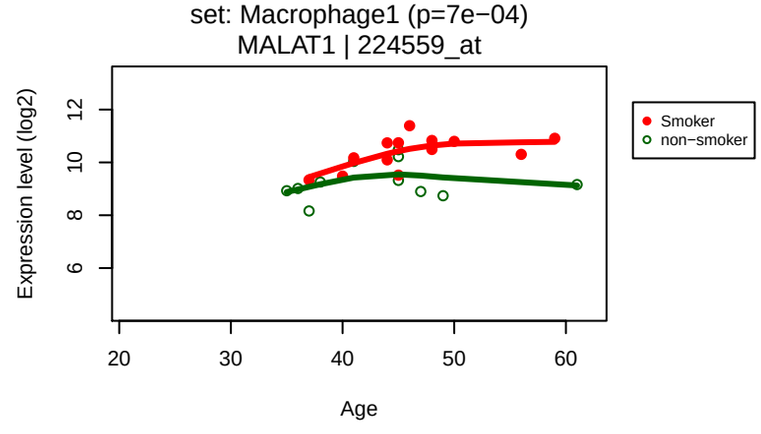

set: SmallAirway1 (p=0.0049)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker

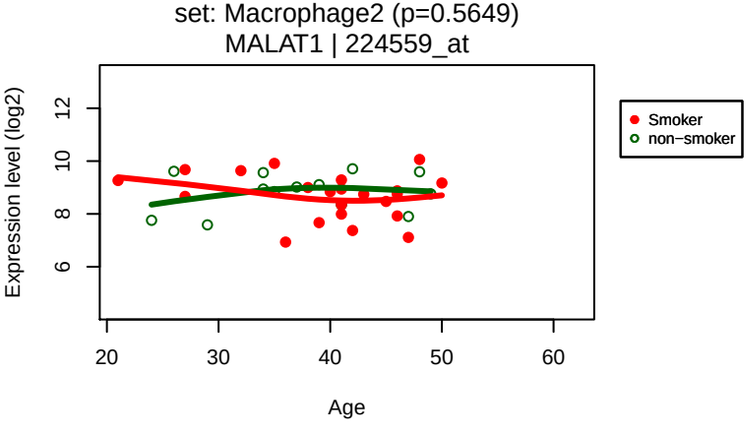

set: SmallAirway2 (p=0.0488)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker

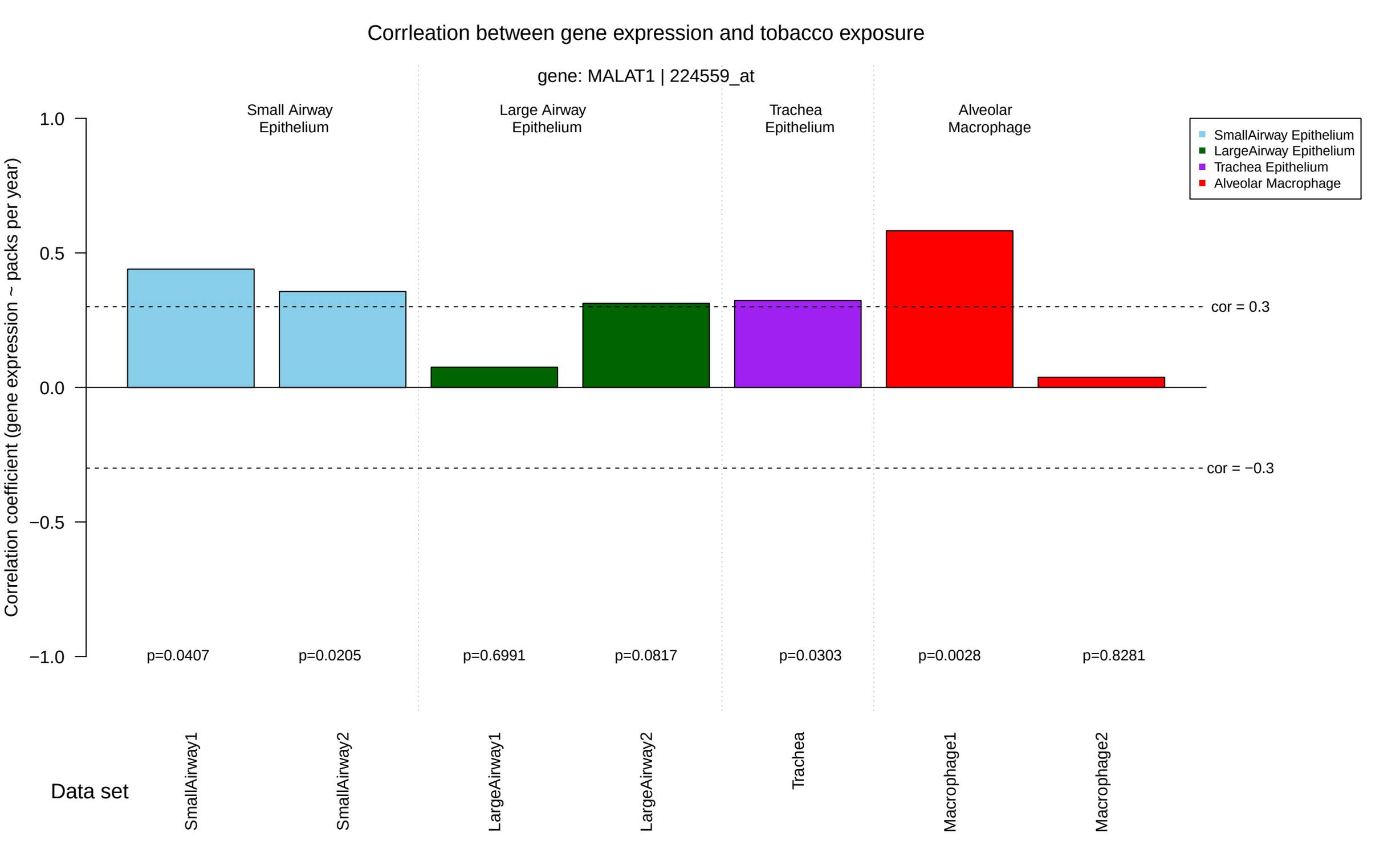

set: LargeAirway1 (p=0.7029)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4441)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0062)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=7e−04)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5649)
MALAT1 | 224559_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MALAT1 | 224559_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0407
p=0.0205
p=0.6991
p=0.0817
p=0.0303
p=0.0028
p=0.8281
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2